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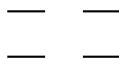
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1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned} a_u^l &= +b_{1g} + a_u \\ b_{1u}^l &= +a_g + b_{1u} \\ b_{2g}^l &= +b_{3u} + b_{2g} \\ b_{3g}^l &= +b_{2u} + b_{3g} \\ a_u^r &= -b_{1g} + a_u \\ b_{1u}^r &= -a_g + b_{1u} \\ b_{2g}^r &= -b_{3u} + b_{2g} \\ b_{3g}^r &= -b_{2u} + b_{3g} \end{aligned}$$

2 Monomer States and Configurations

$$\begin{aligned}
S &= + \begin{pmatrix} \overline{} & \overline{} \\ \downarrow\uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g} \right| \\
Q &= + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g} \right| \\
i^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}} \right| \\
e^3 b_{2u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}} \right| \\
e^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}} \right| \\
i^3 b_{3u} &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}} \right|
\end{aligned}$$

3 Linear Combinations

3.1 $S * Q + Q * S$

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$-2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

3.2 $S * Q - Q * S$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right|$$

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right|$$

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|$$

3.3 $i^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

3.4 $i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

3.5 $i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| \\ +4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| \\ -4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| \\ +4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right|$$

3.6 $i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

3.7 $i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|$$

3.8 $i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

3.9 $i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|$$

3.10 $e^3 b_{2u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

3.11 $e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right| \\ -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

3.12 $e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|$$

3.13 $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \right|$$

3.14 $e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\ -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|$$

3.15 $e^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\ + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\ - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

3.16 $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

3.17 $e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| \\ - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \cdot \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| \\ +4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| \\ -4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| \\ +4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|$$

3.18 $i^3 b_{3u} * i^3 b_{3u}$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right|$$

$$- 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_g
$S * Q - Q * S$	b_{1u}
$i^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$	a_g
$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$	b_{1u}
$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$	a_u
$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$	b_{1g}
$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$	a_u
$e^3 b_{2u} * e^3 b_{2u}$	a_g
$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$	b_{1g}
$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$	a_u
$e^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$	a_g
$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$	b_{1u}
$i^3 b_{3u} * i^3 b_{3u}$	a_g

$\Rightarrow$ 7x a_g and 3x b_{1u} and 4x b_{1g} and 4x a_u

5 CI Vector Conclusion

Dimer States and Their CI Vectors:

- $S * Q + Q * S$:

$$+2 \cdot |0220aaaa| + 2 \cdot |02aaaa20| - 2 \cdot |0a2aa2a0| + 2 \cdot |0aa0a22a| + 2 \cdot |a22a0aa0| - 2 \cdot |a2a00a2a| \\ + 2 \cdot |aa2002aa| + 2 \cdot |aaaa0220|$$

- $S * Q - Q * S$:

$$+2 \cdot |022aaaa0| - 2 \cdot |02a0aa2a| + 2 \cdot |0a20a2aa| + 2 \cdot |0aaaa220| - 2 \cdot |a2200aaa| - 2 \cdot |a2aa0a20| \\ + 2 \cdot |aa2a02a0| - 2 \cdot |aaa0022a|$$

- $i^3 b_{2u} * i^3 b_{2u}$:

$$+ |a2a0a2a0| - 2 \cdot |a2a00a2a| - 2 \cdot |a22a0aa0| + 2 \cdot |aaaa0220| - 2 \cdot |aa2002aa| - 2 \cdot |02aaaa20| \\ + 2 \cdot |0220aaaa| - 2 \cdot |0aa0a22a| - 2 \cdot |0a2aa2a0| + |0a2a0a2a|$$

- $i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$:

$$+ 2 \cdot |a2a0a2a0| - 2 \cdot |0a2a0a2a|$$

- $i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$:

$$+ 4 \cdot |a2aa0a20| - 4 \cdot |a2200aaa| + 4 \cdot |aaa0022a| + 4 \cdot |aa2a02a0| - 4 \cdot |02a0aa2a| - 4 \cdot |022aaaa0| \\ + 4 \cdot |0aaaa220| - 4 \cdot |0a20a2aa|$$

- $i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$:

$$+ 2 \cdot |a2a0aa20| + 2 \cdot |aa20a2a0| - 2 \cdot |a2a002aa| - 2 \cdot |02aaa2a0| - 2 \cdot |0a2aaa20| - 2 \cdot |aa200a2a| \\ + 2 \cdot |0a2a02aa| + 2 \cdot |02aa0a2a|$$

- $i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$:

$$- 2 \cdot |a220aaa0| + 2 \cdot |aaa0a220| + 2 \cdot |a2aa02a0| - 2 \cdot |02a0a2aa| + 2 \cdot |0a20aa2a| - 2 \cdot |aa2a0a20| \\ - 2 \cdot |0aaa022a| + 2 \cdot |022a0aaa|$$

- $i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$:

$$+ 2 \cdot |a2a0aa20| + 2 \cdot |aa20a2a0| + 2 \cdot |a2a002aa| + 2 \cdot |02aaa2a0| - 2 \cdot |0a2aaa20| - 2 \cdot |aa200a2a| \\ - 2 \cdot |0a2a02aa| - 2 \cdot |02aa0a2a|$$

- $i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$:

$$- 2 \cdot |a220aaa0| + 2 \cdot |aaa0a220| - 2 \cdot |a2aa02a0| + 2 \cdot |02a0a2aa| + 2 \cdot |0a20aa2a| - 2 \cdot |aa2a0a20| \\ + 2 \cdot |0aaa022a| - 2 \cdot |022a0aaa|$$

- $e^3 b_{2u} * e^3 b_{2u}$:

$$+ |a2a0a2a0| + 2 \cdot |a2a00a2a| + 2 \cdot |a22a0aa0| - 2 \cdot |aaaa0220| + 2 \cdot |aa2002aa| + 2 \cdot |02aaaa20| \\ - 2 \cdot |0220aaaa| + 2 \cdot |0aa0a22a| + 2 \cdot |0a2aa2a0| + |0a2a0a2a|$$

- $e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$:

$$+2 \cdot |a2a0aa20| + 2 \cdot |aa20a2a0| - 2 \cdot |a2a002aa| - 2 \cdot |02aaa2a0| + 2 \cdot |0a2aaa20| + 2 \cdot |aa200a2a|$$

$$-2 \cdot |0a2a02aa| - 2 \cdot |02aa0a2a|$$

- $e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$:

$$-2 \cdot |a220aaa0| + 2 \cdot |aaa0a220| + 2 \cdot |a2aa02a0| - 2 \cdot |02a0a2aa| - 2 \cdot |0a20aa2a| + 2 \cdot |aa2a0a20|$$

$$+2 \cdot |0aaa022a| - 2 \cdot |022a0aaa|$$

- $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$:

$$+2 \cdot |a2a0aa20| + 2 \cdot |aa20a2a0| + 2 \cdot |a2a002aa| + 2 \cdot |02aaa2a0| + 2 \cdot |0a2aaa20| + 2 \cdot |aa200a2a|$$

$$+2 \cdot |0a2a02aa| + 2 \cdot |02aa0a2a|$$

- $e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$:

$$-2 \cdot |a220aaa0| + 2 \cdot |aaa0a220| - 2 \cdot |a2aa02a0| + 2 \cdot |02a0a2aa| - 2 \cdot |0a20aa2a| + 2 \cdot |aa2a0a20|$$

$$-2 \cdot |0aaa022a| + 2 \cdot |022a0aaa|$$

- $e^3 b_{3u} * e^3 b_{3u}$:

$$+ |aa20aa20| - 2 \cdot |aa2002aa| - 2 \cdot |aaaa0220| + 2 \cdot |a22a0aa0| - 2 \cdot |a2a00a2a| - 2 \cdot |0a2aa2a0|$$

$$+2 \cdot |0aa0a22a| - 2 \cdot |0220aaaa| - 2 \cdot |02aaaa20| + |02aa02aa|$$

- $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$:

$$+2 \cdot |aa20aa20| - 2 \cdot |02aa02aa|$$

- $e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$:

$$+4 \cdot |aa2a02a0| - 4 \cdot |aaa0022a| + 4 \cdot |a2200aaa| + 4 \cdot |a2aa0a20| - 4 \cdot |0a20a2aa| - 4 \cdot |0aaaa220|$$

$$+4 \cdot |022aaaa0| - 4 \cdot |02a0aa2a|$$

- $i^3 b_{3u} * i^3 b_{3u}$:

$$+ |aa20aa20| + 2 \cdot |aa2002aa| + 2 \cdot |aaaa0220| - 2 \cdot |a22a0aa0| + 2 \cdot |a2a00a2a| + 2 \cdot |0a2aa2a0|$$

$$-2 \cdot |0aa0a22a| + 2 \cdot |0220aaaa| + 2 \cdot |02aaaa20| + |02aa02aa|$$